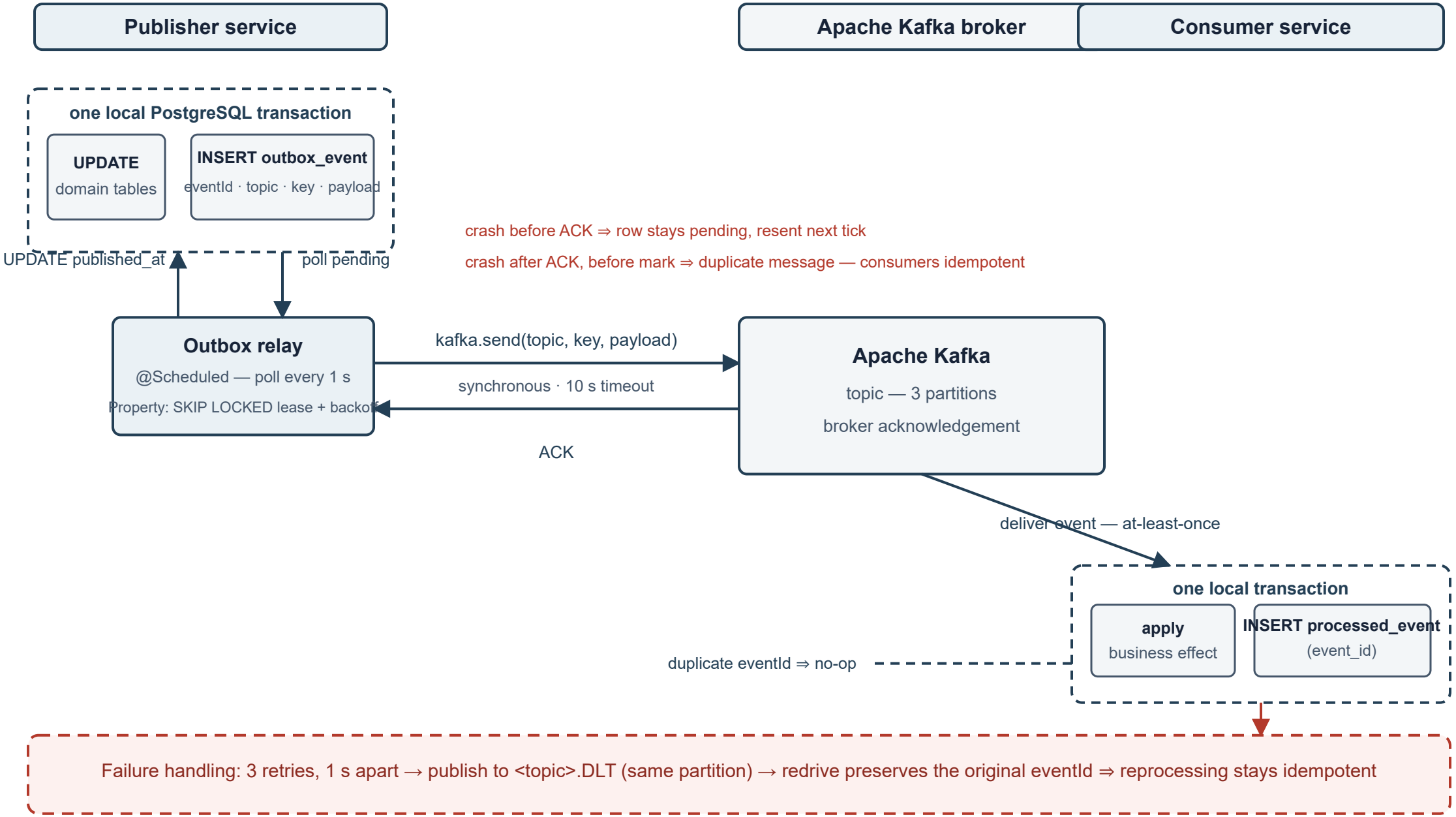

Publisher service
Apache Kafka broker
Consumer service
one local PostgreSQL transaction
UPDATE
domain tables
INSERT outbox_event
eventId · topic · key · payload
UPDATE published_at
poll pending
crash before ACK ⇒ row stays pending, resent next tick
crash after ACK, before mark ⇒ duplicate message — consumers idempotent
Outbox relay
@Scheduled — poll every 1 s
Property: SKIP LOCKED lease + backoff
kafka.send(topic, key, payload)
synchronous · 10 s timeout
ACK
Apache Kafka
topic — 3 partitions
broker acknowledgement
deliver event — at-least-once
one local transaction
apply
business effect
INSERT processed_event
(event_id)
duplicate eventId ⇒ no-op
Failure handling: 3 retries, 1 s apart → publish to <topic>.DLT (same partition) → redrive preserves the original eventId ⇒ reprocessing stays idempotent